

Zero Peak Homes: Real Homes, Real Savings

by **Ryan Kerr**

A net-zero energy home (ZEH) has become a familiar construct to most *Home Energy* readers. But what we at the Building Industry Research Alliance (BIRA) have learned as we have worked to make this concept a reality has piqued our interest in another type of home—the zero peak home (ZPH). While ZEHs create as much energy as they use over the course of a year, ZPHs draw no grid electricity during peak hours. These homes offer significant benefits to their occupants. They could also help our country to considerably reduce peak electricity demand and overall carbon emissions.

BIRA is one of six teams working with the U.S. Department of Energy's (DOE) Building America program and national labs to create cost-effective, marketable ZEHs by 2020. Based in California, BIRA has worked with regional and national builders to construct over 10,000 Building America homes in the past five years. Today's near ZEHs reduce energy use and utility bills by at least 60% compared to an average home built today, through the use of energy efficiency and solar energy technologies. Even with these added features, these ZEHs remain cost competitive with traditional homes. When we add together all of the economic contributions that communities of ZEHs deliver—including environmental benefits, marketing and sales advantages for their builders, and cost savings for electric utilities—the economic picture is even rosier. The ability to reduce peak electricity demand and carbon

emissions is one of the most significant and promising of these contributions.

Why Focus on Peak?

Producing the extra electricity needed to meet peak demand is expensive—both financially and environmentally. Large capital investments are needed to provide the electrical generation, transmission, and distribution capabilities required to meet peak electricity demand. But since demand is variable, these large capital investments may lie idle or underused for much of the day. Indeed, the electricity infrastructure is oversized and used inefficiently most of the year, with utility load factors hovering around 60%. Taking all these factors into account, the marginal cost of peak electricity can be more than 10 times that of off-peak electricity.

The situation is even worse when we consider environmental costs. Almost all electricity use produces carbon dioxide (CO₂) emissions. But peak electricity use can be even more carbon intensive than off-peak use. There are several reasons why. First and perhaps most important, peak use requires the construction of additional generation, transmission, and distribution capacity. This construction eats up land and energy, as newly fabricated construction materials are transported to the new facilities. Not only is more carbon emitted in these processes, but trees and other carbon sinks are displaced by the added infrastructure, such as substations and distribution lines.

In addition to the costs and environmental impacts of an oversized

electric infrastructure, there is the pollution caused by marginal generation, or peaker plants. These plants produce electricity only during the highest demand periods. As these plants are seldom used, their efficiency is financially less important to their operators than the efficiency of baseload plants, which operate much more frequently. Peak electricity suppliers cannot justify the added capital cost of improved efficiency or emission controls when the payback would outrun the life of the plant. Peaker plants are typically simple combustion turbines with generators. The exhaust heat is not used to produce additional electricity, as it is in their more sophisticated counterparts, which makes them even less efficient. Similarly, peaker plants seldom have advanced emissions controls, because exhaust gases in peaker plants run very hot, and advanced controls for these hot gases would be prohibitively expensive. The typical peaker plant burns oil or natural gas, but some peaker plants burn diesel fuel, which is more polluting than either.

The Sacramento Municipal Utility District (SMUD), which supplies electricity to the Sacramento, California, metropolitan area, recently conducted an analysis of its generation resources. This analysis concluded that the average peaker plant in the SMUD generation system emits almost 40% more CO₂ per megawatt produced compared to baseload generation. This increase represents only emissions from generation; it does not include the embodied energy, and carbon, inherent in the peaking infrastructure.



Concrete Structural Insulated Panel (CSIPs) combine the tightness and insulation value of traditional SIPs with additional mass.

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CLARUM HOMES

This Clarum Homes desert prototype house, which was built with T-MASS walls, maintained indoor comfort on very hot test days, even when the air conditioning was shut off during the afternoon.

Average 15 Minute Interval Peak Demand ZEH vs. Non ZEH July, 2005

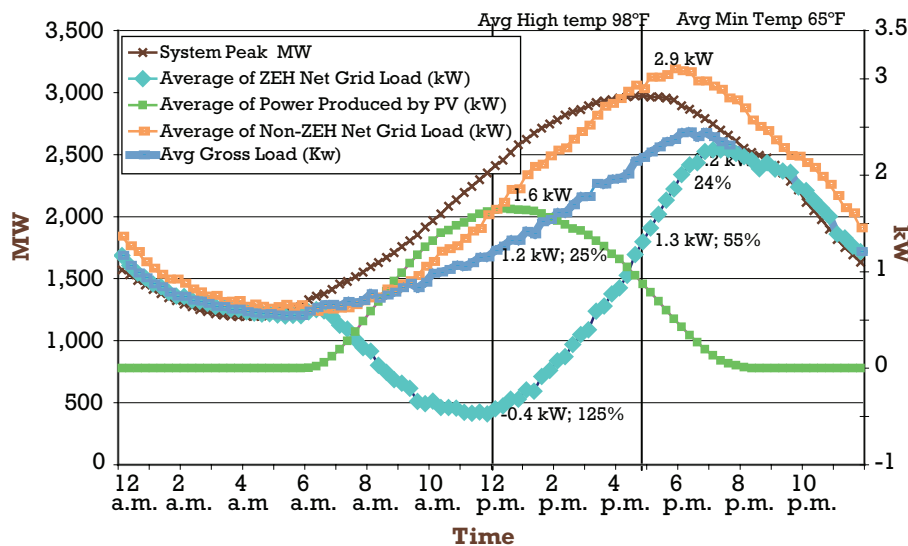


Figure 1. SMUD's analysis of time-of-use (TOU) power consumption at Premier Gardens has shown real and lasting reductions during peak periods, as compared to consumption at a neighboring community of similarly sized homes built and monitored during the same time period.

Where To Start?

Nationwide, air conditioning accounts for one-third of the overall electrical load during hot summer afternoons when demand is highest. In California, commercial and residential air conditioning accounts for 29% to 40% of statewide peak electricity demand. Residential air conditioning represents the greatest portion of peak demand, with commercial air conditioning second and commercial lighting third. Due to the distributed

nature of residential housing, the residential market has been a relatively untapped resource for demand-side management, compared to the commercial and industrial markets. However, new research suggests that residential consumers can even out a utility's overall load by reducing peak electricity use and generating solar-electric power on-site. On a community scale, ZPHs could have a substantial impact on peak demand, and by doing so could benefit the

environment, the utility, and the homeowner.

For all of these reasons, BIRA has begun working toward designing zero peak homes. The basic design philosophy is, first, to shed as much peak load as possible by implementing cost-effective energy efficiency measures; second, to utilize west-facing solar generation; and finally, to reduce or eliminate the use of air conditioning during peak hours. Given the impact that air conditioning has on peak use, shifting this use to off-peak is the most important and perhaps the most challenging piece of the puzzle.

Near Zero Lessons

The Premier Gardens near zero energy community in Sacramento is a fairly well-known and well-monitored housing development that has been an excellent workshop for efficient building techniques. The houses in this community combine significant energy efficiency features with photovoltaic (PV) solar-electric generation, both of which help to reduce peak demand. SMUD's analysis of time-of-use (TOU) power consumption at Premier Gardens has shown real and lasting reductions during peak periods, as compared to consumption at a neighboring community of similarly sized homes built and monitored during the same time period (see Figure 1).

In July 2005, summer temperatures set a new record in this community with average daily highs of 98°F and average lows of 65°F. SMUD also set a new record for system peak demand at 5 pm on July 15, 2005, exceeding the previous system high by 5%. During this month, SMUD's ongoing monitoring of the average 15-minute interval peak demand at Premier Gardens and the control community revealed peak reduction findings that were notable in several ways. First, the shape of the demand curve for the non-ZEHs (orange curve in Figure 1) closely mirrors the overall load shape of SMUD's system peak (the brown curve in Figure 1), highlighting the

Trading Up to a Hybrid Home

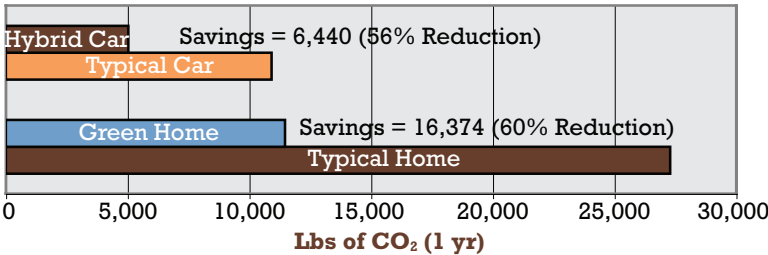
More and more individuals and families are trying to reduce their personal carbon footprint. With so many options, what are the best techniques for achieving this goal?

Well, let's start with the big picture: Where do our carbon emissions come from? In the United States, they come from four major sectors: industrial, residential, commercial, and transportation. According to Energy Information Administration data and Architecture2030 calculations that consider the embodied energy from construction and materials as well as the energy used to operate the building, emissions for buildings across all sectors account for 48%, or almost half, of all carbon emissions nationwide. By comparison, emissions from the transportation sector account for roughly 27%. The residential sector accounts for the highest proportion of emissions from buildings; more than one-fifth of all U.S. carbon emissions come from the operation of residential buildings alone.

So what does this mean for our personal carbon footprint? While trading up to a hybrid car is a great option, trading up to an energy-efficient home with solar might be even better. Let's compare the average carbon dioxide (CO₂) savings realized by switching to a hybrid car to the savings realized by moving into a green home (see Figure A). While the percentage of CO₂ emissions saved is almost the same in both cases (56% versus 60%), the carbon impact of operating a home is significantly higher than that of driving a car. Therefore the net savings from moving into a green home will be significantly higher than the savings from switching to a hybrid car. Consequently, a family with a typical home and two hybrid cars has a larger carbon footprint than one with a green home and two typical cars.

Today, builders across the country are beginning to endorse this idea by building homes that use 60% less energy than typical homes and by selling them at prices that match the prices charged by their nongreen competitors. While this is good news for consumers, builders are also finding that they benefit by building green; green homes mean faster sales. Builders are able to invest additional money in green features—such as double-paned, low-e windows and solar-electric systems—by marketing the benefits to consumers and selling homes faster (see “SolarSmart Homes Sell Faster,” p. 10). Faster sales mean lower project carrying costs and more profit. Even if a builder passes some or all of the cost on

Figure A. Carbon Footprint of Your Home and Car



to the home buyer, a cost-effective green home creates positive monthly cash flow for the homeowner, who sees the savings in utility costs offset the additional mortgage cost associated with the green features. Even with rising gas prices, a hybrid car may not meet this cost-neutral criterion.

So why have we gravitated toward hybrid cars as opposed to “hybrid” homes? One likely reason is that our homes don’t have tailpipes. It is easy to visualize the carbon impact of our car— we can actually see the exhaust. With a home, our power is typically generated miles away and we don’t see the emissions that result from turning on our TV or cranking the thermostat down in the summer. In addition, hybrid cars are often equipped with a digital display showing real-time fuel consumption, while our homes are not; however, this is starting to change. For less than \$200, you can buy a meter that shows you the real-time electricity consumption of your home, and in many cases the carbon impact of that consumption. These “digital tailpipes” have been shown to reduce a home’s energy bills—in some cases up to 10%—when the occupants use them to understand and manage their consumption.

From creating national policy to making individual buying decisions, a home’s energy consumption and the carbon impact of that consumption, are important factors influencing global warming and ought to be recognized as such. So when you spend your carbon- fighting dollars, take a second look at a “hybrid” home.

To learn more about the architecture2030 challenge, visit www.Architecture2030.org.

For an inexpensive in-home power meter you can install yourself, visit www.bluelineinnovations.com/powercostmonitor.php.

significance of residential demand at peak. At 5 pm, the ZEH community’s demand was 56% below that of the non-ZEH community. Interestingly, roughly half of the peak load reduction was a result of efficiency measures, which were implemented at one-fifth the cost of the roof-mounted PV solar-electric panels. Many of the ZEHs feature east-facing solar panels, which produce most of their power during the late morning and early afternoon; west-facing panels produce most of	their power during the late afternoon, coinciding with most utilities’ superpeak hours. With east-facing panels, PV electricity production (the green curve in Figure 1) peaks around noon, rather than later in the day. The PVs could have been oriented toward the west and south in most cases, if peak production had been the primary goal. An analysis was done to determine the impact of having all west-facing PVs on Premier’s near-ZEHs. This analysis showed that, had the Premier Gardens development	been designed to maximize west-facing PVs, the net grid load could have been shifted almost three hours. In addition, the average demand at 5 pm would have been reduced from 1.3 kilowatts, as built, to 0.75 kilowatts with all west-facing PVs. BIRA found that if PV systems had been allowed on the front of the homes, the as-built development could have eliminated all but 4 of the 23 homes with east-facing PVs. Finally, other roof types could have been used to eliminate all east-facing PVs. However, even without this
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This prototype home in Borrego Springs, California, was built with T-MASS walls, which have layers of concrete and Styrofoam.

optimization, this ZEH community clearly demonstrates a significant reduction in peak demand.

Hot-Dry Climates Present Opportunities

Fortunately, many Western climates have attributes that make building ZPHs a realistic goal. The relatively low humidity makes it possible to use evaporative coolers to replace air conditioning. Direct evaporative coolers simply evaporate water into the air, causing a cooling effect. They tend to be simple and inexpensive, and they use much less energy than traditional air conditioning. They do have some drawbacks—they increase indoor humidity; they don't cool the air as well as air conditioning on very hot or humid days; and they increase water usage, which can be a drawback in drought-prone areas. However, the new two-stage and indirect evaporative coolers draw about as much electricity as their direct counterparts, and they do not increase indoor humidity. They do come with a higher price tag than direct models.

In Western climates, hot summer days are often followed by relatively cool nights. This makes night ventilation a good option, especially during the swing cooling seasons

of late spring and early fall. There are new ventilation systems, such as NightBreeze and SmartVent, that automatically draw in cool night air and distribute it through the home's air handler and ductwork. These systems provide benefits that whole-house fans cannot match. They filter outside air for pollen and pollution; they operate automatically; and they reduce security risks, since they can operate with the windows closed. Night ventilation, however, does little to reduce demand during the peak hours of late afternoon, when houses heat up and require additional cooling.

Having considered all of the requirements of a ZPH, Clarum Homes, in partnership with Davis Energy Group, BIRA, ConSol, and the National Renewable Energy Laboratory (NREL), built four prototype homes in the harsh desert climate of Borrego Springs in Southern California. Two of the houses were built with T-MASS walls. These walls have 4 inches of concrete on the inside, 4 inches of Styrofoam in the middle, and 2 inches of concrete on the outside. The third house was built with structural insulated panels (SIPs), with 4 inches of rigid foam sandwiched between two pieces of oriented strand board (OSB). The fourth house was built with 2 x 6 wood framing and spray foam insulation. Two of the

houses have two-stage evaporative cooling systems; one has an air conditioner with an evaporative-cooled condenser; and one has a 20.5-SEER air conditioner. All four houses were tested throughout a summer of unoccupied use. Each house performed well, but they all performed differently.

After considering overall efficiency, current building practices, and cost, Clarum Homes chose the SIP wall system as being the best suited for future projects. SIPs expedite wall construction; they provide excellent insulation value; and they create a very tight envelope when installation protocols are followed. However, the T-MASS walls performed exceptionally well during the peak-shifting experiment described below.

All the houses were reasonably tight and well insulated, and all of them had extended overhangs. These features suggested that the houses could be precooled during the summer and allowed to coast during peak hours, in order to shift cooling energy consumption off-peak. The houses were precooled to nearly 72°F by 12 noon. The cooling equipment was then shut off and the houses were allowed to coast. During the day of the experiment, the outside temperature reached 105°F. The T-MASS house coasted through the day with an internal air temperature rise of only 3.5°F. This outperformed the SIP and stick-built houses, which rose 8°F and 9°F respectively over the same period. Each house was able to shift all cooling energy use to off-peak, but T-MASS was the only wall system that was able to maintain indoor comfort. While this result was very promising, the T-MASS walls were certainly not a panacea for shifting cooling in production housing; the T-MASS walls were much more expensive than the 2 x 6 walls or the SIPs, and they were difficult to procure and erect. Furthermore, total electricity use for precooling was higher in the T-MASS house than it was in the other houses, because there was more internal mass to charge.



RYAN KERR



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CSIPs look like traditional SIPs except that both of the OSB pieces are substituted with concrete wallboard, similar to a cement tile backer.

On to Fresno

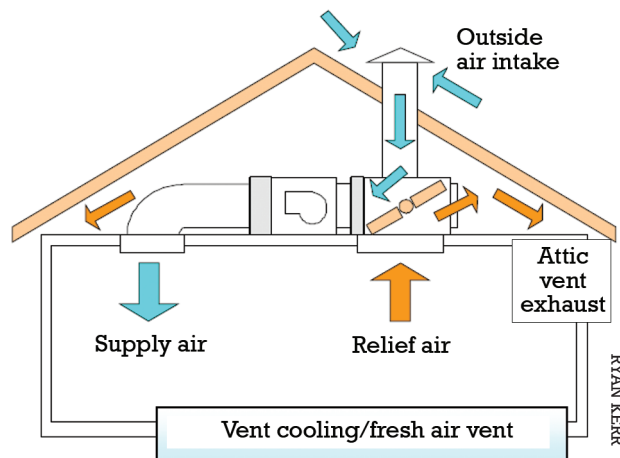
The next step in BIRA's journey toward ZPHs brought our team to Fresno, California, where innovative building partner Alvis Projects is in the process of constructing three affordable green homes as part of the city of Fresno's Green Building Demonstration Project. Given the success of the T-MASS walls in facilitating precooling, and the practicality and low cost of the SIPs, BIRA moved to a new wall system for the second Fresno home. Concrete SIPs (CSIPs) combine the tightness and insulation value of traditional SIPs with additional mass. CSIPs look like traditional SIPs except that both of the OSB pieces are substituted with concrete wallboard, similar to a cement tile backer. While walls constructed with CSIPs lack the mass of the T-MASS wall, the $\frac{1}{2}$ inch of concrete wallboard on the inside surface,

coupled with $\frac{5}{8}$ -inch gypsum board on top of that, together with concrete floors and countertops, does create a substantial amount of interior mass.

The CSIP wall system will be combined with a NightBreeze and a small supplemental air conditioning system. The NightBreeze will be used to draw in cool night air, drawing heat out of the thermal mass to prepare the home for precooling, and to reduce the need for air conditioning. Except during the hottest times of the year, the NightBreeze, assisted by the thermal mass, insulation, and tight envelope, should make it unnecessary to use the air conditioning at all. We decided not to use a two-stage or indirect evaporative cooler, because it might not cool the home effectively throughout the year and would therefore require an air conditioner for backup. Given that the cooling load is small and the home will be precooled by the NightBreeze, the peak demand and overall electricity savings from an evaporative cooler would not justify the cost of installing one. To utilize the mass in the heating season, a clerestory was added. This clerestory will allow the sun's energy to heat the interior mass in the winter and will provide daylighting all year long. When construction on these homes is complete, they will be heavily instrumented to monitor the wall system's influence on heating and cooling energy use.

The Challenges of Shifting Cooling Loads

BIRA and the building industry are learning how to design and build homes that facilitate precooling and substantially reduce peak use. But persuading homeowners to shift use to off-peak is still a challenge. NightBreeze can save the homeowner money by providing a cheaper



The NightBreeze Integrated Night Ventilation Cooling System is used to draw in cool night air, drawing heat out of the thermal mass to prepare the home for precooling, and to reduce the need for air conditioning.

alternative to air conditioning, but promoting precooling in the absence of favorable TOU rates is difficult. Most utility TOU rates are not structured in such a way as to make it pay for the homeowner to shift loads; peak periods are often too long to be practical for the homeowner, and the difference between peak and off-peak rates is not large enough to compensate for the inconvenience. However, ZPHs are perfectly suited to take advantage of well-designed TOU rates; their greatly reduced demand, coupled with solar-electric generation during peak periods, displaces almost all net grid consumption to cheaper off-peak periods. Simple programmable thermostats are available that make precooling easy to automate, but until TOU rates are restructured in a way that benefits the homeowner, it is unlikely that ZPHs will fulfill their potential. BIRA will continue to work with utility and builder partners throughout the West to meet our goal of building and marketing cost-effective ZPHs that reduce energy consumption by at least 60% and consume no grid electricity during peak hours.

Investing in Peak Reductions

One of the techniques that we use in designing cost-effective ZEHs is to minimize the size of the air

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conditioning compressor. Traditionally, air conditioners are sized to meet cooling demands on the hottest days of the year. Much of this capacity will go unused the remainder of the year, causing increased cycling and inefficient use. With better insulation, windows, shading, and tight, well-insulated ducts, compressors can be halved in size and still deliver ample cooling for indoor comfort. The money saved from moving to a smaller compressor offsets the costs of the features that allowed the compressor to be made smaller in the first place.

This concept can be applied to the electrical system as a whole. If ZPHs were commonplace, peak electricity demand would be greatly reduced and there would be less need for larger transmission and distribution equipment, and less need to increase generating capacity. The capacity of the system could also be designed with tighter tolerances, because the

range of loads it would experience would be narrower. The cost savings could then be used to construct more efficient power plants with better emissions controls. Or, as with zero energy homes, the savings could be invested in the thing that produced the savings in the first place—more zero peak homes—by giving an upfront incentive to builders or buyers. TOU rates that sufficiently benefit load shifting could provide another such incentive.

Reducing peak electricity demand is financially practical for the cost savings it provides and environmentally important for the emissions reduction it creates. Implemented on a large scale, zero peak communities could transform the electric grid and create a cleaner, more efficient electrical infrastructure. This can happen only if there is continued research and investment on the part of those who would benefit

the most: the electricity industry. Given the tremendous impact of residential peak demand on the electric grid, particularly with regard to air conditioning, it is essential that the electricity and building industry work together to pursue the goal of building more ZPHs. Now more than ever, reducing peak electricity consumption is imperative, and zero peak homes are on the forefront of this effort.

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For more information:

For more information on any of the projects mentioned in the article, visit www.BIRA.ws or contact Ryan Kerr at rkerr@consol.ws.